

Module : 4

- 1) Explain working of Prandtl pitot static tube with sketch.
- 2) Explain working of:
 - ~~a)~~ velocimeter
 - ~~b)~~ Schlieren apparatus
 - ~~c)~~ Interferometer
 - ~~d)~~ Hot wire anemometer} with figures.
- 3) Is it desirable to locate the shock wave at the diffuser throat during the operation of a supersonic wind tunnel? Justify your answer.
- 4) What are various types of wind tunnels used for low ~~pressure~~ and high speed testing of models? Describe briefly the special problems of supersonic wind tunnel.
- 5) Explain temperature recovery factor.
- ~~6)~~ Explain working principle of shadowgraph with figure.
- 7) Explain shock tube working with figure.
- 8) Differentiate between blower and suction type subsonic wind tunnels with figure.
- 9) Explain why the diffuser throat area is larger than nozzle throat area in a supersonic wind tunnel.
- ~~10)~~ Explain how closed circuit wind tunnels are advantageous over open circuit wind tunnels for supersonic testing.
- 11) How is it possible to employ Pitot tube for measurement of Mach no. in a supersonic stream?

12) Explain with figures:

- a) High velocity total temperature probe
- b) Constant temperature hot wire anemometer
- c) Constant current hot wire anemometer

13) Explain supersonic wind tunnel with fig.

14) What is significance of Rayleigh supersonic pitot formula?